Primary 3 · Science

Science · Set 4 — Answer Key

80 marks · 1 h 15 min · Focus practice

For parents. Give this to the pupil only after the paper is finished. Written answers: accept any answer with the same meaning, in a complete sentence with correct tense.

Section A: Multiple-Choice Questions (40 marks)

No.	Answer	Marks
1	(1) The amount of water given	2
2	(1) The amount of sunlight and the type of soil	2
3	(2) Plant 2 (medium water)	2
4	(1) The amount of water given to each plant	2
5	(3) Adult plant with flowers and fruits	2
6	(3) Sunlight	2
7	(4) Icicle	2
8	(4) Birds	2
9	(4) Dog	2
10	(1) Yeast	2
11	(4) Cockroach	2
12	(3) 4	2
13	(2) Pupa	2
14	(2) It still has a tail and is starting to breathe air	2
15	(4) Clear glass or plastic	2
16	(3) A small iron key	2
17	(2) A rubber band	2
18	(2) Cactus	2
19	(1) Frog	2
20	(3) Fungi can reproduce using spores	2

Section B: Structured Questions (16 marks)

No.	Answer	Marks
21(a)	False	1
21(b)	True	1
22	(1) How high each ball bounces	2
23	Waterproof (it does not let water leak or soak through) — and strong enough to hold the weight of the water.	2
24	Transparent (clear / see-through), so customers can see the food through it.	2
25	(2) A patch of damp, decaying leaves	2
26(a)	False	1
26(b)	True	1
27(a)	True	1
27(b)	False	1
28	Food (or nutrients) and water.	2

Section C: Open-Ended Questions (24 marks)

No.	Answer	Marks
29	Only the number of times the water was stirred was changed. The amount of sugar, the amount of water, and the water's starting temperature were kept the same for every cup. This makes it a fair test because any difference in how fast the sugar dissolves is caused only by the amount of stirring, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the number of stirs), 1 mark for correctly stating what was kept the same (amount of sugar, amount of water, starting temperature), 1 mark for explaining why this makes the test fair. | Hint: Use the frame: Only the ___ was changed, so any difference in ___ is caused by ___.	3

30	Only the height of the ramp was changed. The toy car used and the floor it rolled onto were kept the same for every release. This makes it a fair test because any difference in how far the car rolls is caused only by the height of the ramp, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the height of the ramp), 1 mark for correctly stating what was kept the same (the toy car, the floor), 1 mark for explaining why this makes the test fair. | Hint: Use the frame: Only the ___ was changed, so any difference in ___ is caused by ___.	3
31	She could gently touch the caterpillar to see if it curls up or moves away, and she could watch closely to see if it is breathing or slowly moving any part of its body. — Marking guide: 1 mark for a sensible test of response to touch/stimulus, 1 mark for a second sensible test (e.g. checking for breathing or slow movement), 1 mark for a clear, complete explanation. Give 0 marks for actions that are CARE rather than a TEST (e.g. giving it a leaf to eat, or moving it to a safer spot). | Hint: Think of two things you could DO to the caterpillar or WATCH it do, not ways of caring for it.	3
32	Mosquito eggs hatch into larvae that live in water, and the larvae grow into pupae, which also stay in the water, before becoming adult mosquitoes. Without stagnant water for the eggs, larvae and pupae to develop in, fewer mosquitoes can grow into adults, so there is less chance of dengue spreading. — Marking guide: 1 mark for noting that mosquito eggs/larvae/pupae need water to develop, 1 mark for linking removing water to fewer mosquitoes growing into adults, 1 mark for a clear overall explanation connecting this to preventing dengue. | Hint: Think about which stages of the mosquito's life cycle happen in water.	3
33	A tadpole starts life breathing through gills and swimming with a tail. As it grows into a frog, it grows four legs, loses its tail, and its gills are replaced by lungs for breathing air. A chick already looks similar to a hen and mostly just grows bigger, without losing a tail or changing how it breathes. — Marking guide: 1 mark for describing a specific tadpole body change (e.g. gills to lungs, losing the tail, or growing legs), 1 mark for noting the chick does not go through this kind of change, 1 mark for a clear comparison. | Hint: Compare BOTH animals: say what happens to one, then what happens to the other.	3

34	Wood is a good choice because it is strong, so it does not snap easily when gripped firmly. It also does not get slippery when wet, so it is comfortable and safe to hold even in the rain. — Marking guide: 1 mark for stating wood is strong, 1 mark for a second correct property (e.g. does not get slippery, comfortable grip), 1 mark for a clear overall explanation linking the properties to why wood suits an umbrella handle. | Hint: Use the frame: ___ is suitable because it is ___, so ___.	3
35	Although both mushrooms and grass can be found growing on the ground, the key difference is that grass, like other plants, can make its own food using sunlight, but mushrooms, which are fungi, cannot make their own food — they get their food by feeding on dead or decaying matter instead. — Marking guide: 2 marks for stating the key difference clearly for BOTH sides — fungi like mushrooms cannot make their own food, but plants like grass can make their own food using sunlight (1 mark for each side). 1 mark for noting that growing in the same place does not make them the same kind of living thing. | Hint: Where they grow is not the key difference — think about how each one gets its FOOD.	3
36	A seed needs water and air to germinate, but it does not need light to germinate. Since Pot B still received water and air, Seed B was able to germinate and sprout even though it was kept in the dark. Light only becomes important later, for the young plant to grow well and make its own food. — Marking guide: 1 mark for stating that germination needs water and air, not light, 1 mark for linking this to Seed B still germinating because it had water and air, 1 mark for the missing link — noting that light only matters later for growth, not for germination itself. | Hint: Germination and growing afterwards need different things — which one needs light?	3